

PI-VOC



Applications

- Pharmaceutical industry
- Power plants
- Chemical industry
- Air quality
- Refineries and petrochemical plants
- Solvent recovery systems
- Waste water treatment plants
- Marine and off-shore oil wells
- Pulp and paper industry
- Painting and coating operations

Description

PI-VOC Sensors non-intrusive "Smart" sensor that use photo-ionization detector technology to monitor VOC gases over ranges from 0-10 ppm to 0-1,000 ppm.

In the PI-VOC sensor design, we have successfully converted the sensitive laboratory style PID technology into a rugged, environmentally insensitive, and long-life industrial gas sensor. The plug-in, field replaceable detector features over-sized gold-plated connections that help prevent corrosion.

The PI-VOC's rugged framework includes an electro-polished 316 stainless steel housing with fully encapsulated electronics and dual layer surge protection.

This innovative design virtually eliminates sensor failure due to water ingress, corrosion, vibration, and transient spikes.

A primary feature of the Model PI-VOC is embedded intuitive software that simplifies operator interface by guiding the user through routine calibration, configuration, and fault diagnostic functions using a built-in alpha/numeric display.

The PI-VOC is equipped with standard analog 4-20mA and HART outputs.

Features

Failsafe User-Friendly Interface

- LED Display (With Antiglare Cover)
- Full Text Display Method
- Non-intrusive Interface
- Auto Zero/Auto Span
- Pre-emptive Fault Diagnostics

Environmentally Bulletproof

- Electropolished 316SS Construction
- 100% Epoxy Encapsulated Circuitry
- Bulletproof I/O Protection
- Water-proof, Corrosion-Proof, Vibration-Proof

Modular and Serviceable

- Modular Design
- Plug and Play Components
- Quick Thread Release (For Sensor Replacement)
- Integral Calibration Port

System specifications

Sensor Type

Continuous diffusion/adsorption
Photo Ionization Detector (PID)
Plug-in replaceable intelligent type (with replacement lamp)

Sensor Life

2 years typical

Measurement Range

Gas Dependent, ranges as low as 0-10 ppm
or as high as 0-1,000ppm (consult Detcon)

Accuracy/Repeatability

±10% of reading or ±2% of range (greater of)

Response Time

T50 <30 seconds; T90 <60 seconds

Outputs

Linear 4-20 mA DC
HART

Electrical Classification

Explosion proof
CSA and US (NRTL)
Class I, Division 1, Groups A, B, C, D T4 (Tamb = -20°C to +55°C)
Class I, Zone 1, Group IIC, T4
IECEX

Ingress Protection

NEMA 4X, IP66

Safety Approvals

cCSAus

Warranty

PLUG-IN DETECTOR - 1 YEAR
TRANSMITTER - 2 YEARS

Environmental specifications

Operating Temperature Range

-4°F to +122°F; -20°C to +55°C

Storage Temperature Range

-4°F to +122°F; -20°C to +55°C

Operating Humidity Range

0-100% RH non-condensing

Order Guide

PN KSA-9H7-PB4520-010C/ KSA-PM4520-01KC Sensor Assembly with
HART and 316 SS J-Box

* Specifications subject to change without notice

Mechanical specifications

Dimensions

7"H x 2.2" Dia.; 178mmH x 65mm Dia. (sensor assembly only)
11"H x 6.1"W x 3.75"D; 280mmH x 155mmW x 96mmD (with junction box)
Mounting holes (J-box) 5.5"; 140mm center to center

Weight

2 lbs; 0.907kg (sensor only)
6 lbs; 2.72kg (w/aluminum j-box)
9 lbs; 4.08kg (w/stainless steel j-box)

Electrical specifications

Power Input

11 - 30 VDC

Power Consumption

Maximum = 65mA (1.6 watts)

Inrush Current

1.67A @ 24V

RFI/EMI Protection

Complies with EN50270

Analog Output

Linear 4-20mA DC (1,000 ohms max loop load @ 24VDC)
0mA All Fault Diagnostics
2mA In-Calibration
4-20mA 0-100% full-scale
22mA Over-range condition

Status Indicators

4-digit LED display with gas concentration
Full-script menu prompts for AutoSpan,
Set-up Options, and Fault Reporting

Faults Monitored

Loop, Input Voltage,
Zero, Missing Sensor,
Processor, Memory, Calibration

Cable Requirements

Power/Analog
3-wire shielded cable
Maximum distance is 13,300 feet with 14 AWG

I/O Protection

Over-voltage, Miswiring, EMI/RFI Immunity

Integration Options

Remote Alarm Module (Remote Operation and 2 Alarm Relays plus Fault)
Hart Integration Module (Hart Communication Protocol version 7.0, HART Registered)



Copyright © 2025 - GF-30613A-EN. All rights reserved.



Industrial Detection Solutions Company (IDS)
F11 & F12, Dammam 913 Street,
Building No 2857, MODON-2
Dammam Industrial City 2,
Dammam 34325, Saudi Arabia
+966 13 834 9537 | +966 53 968 8090
www.ids.com.sa
manu@ids.com.sa | info@ids.com.sa



IECEx Certificate of Conformity

INTERNATIONAL ELECTROTECHNICAL COMMISSION IEC Certification System for Explosive Atmospheres

for rules and details of the IECEx Scheme visit www.iecex.com

Certificate No.:	IECEx DEK 24.0039X	Page 1 of 4	<u>Certificate history:</u>
Status:	Current	Issue No: 1	Issue 0 (2025-01-29)
Date of Issue:	2025-07-11		
Applicant:	Teledyne Gas & Flame Detection 14880 Skinner Road Cypress TX 77429 United States of America		
Equipment:	Toxic Gas or Oxygen Deficiency Sensor, VOC Gas Sensor		
Optional accessory:			
Type of Protection:	Ex db ib		
Marking:	Ex db ib IIC T4 Gb		

Approved for issue on behalf of the IECEx
Certification Body:

R. Schuller

Position:

Certification Manager

Signature:
(for printed version)

Date:
(for printed version)

2025-07-11

1. This certificate and schedule may only be reproduced in full.
2. This certificate is not transferable and remains the property of the issuing body.
3. The Status and authenticity of this certificate may be verified by visiting www.iecex.com or use of this QR Code.



Certificate issued by:

DEKRA Certification B.V.
Meander 1051
6825 MJ Arnhem
Netherlands





IECEx Certificate of Conformity

Certificate No.: **IECEx DEK 24.0039X**

Page 2 of 4

Date of issue: 2025-07-11

Issue No: 1

Manufacturer: **Teledyne Gas & Flame Detection**
14880 Skinner Road
Cypress TX 77429
United States of America

Manufacturing
locations: **Teledyne Detcon, Inc.**
14880 Skinner Road
Cypress, TX 77429
United States of America

**Industrial Detection Solutions
Company (IDS)**
F11 & F12, Dammam 913 Street,
Building No. 2857
MODON-2
Dammam Industrial City 2 34325
Saudi Arabia

This certificate is issued as verification that a sample(s), representative of production, was assessed and tested and found to comply with the IEC Standard list below and that the manufacturer's quality system, relating to the Ex products covered by this certificate, was assessed and found to comply with the IECEx Quality system requirements. This certificate is granted subject to the conditions as set out in IECEx Scheme Rules, IECEx 02 and Operational Documents as amended

STANDARDS :

The equipment and any acceptable variations to it specified in the schedule of this certificate and the identified documents, was found to comply with the following standards

IEC 60079-0:2017 Explosive atmospheres - Part 0: Equipment - General requirements
Edition:7.0

IEC 60079-1:2014 Explosive atmospheres - Part 1: Equipment protection by flameproof enclosures "d"
Edition:7.0

IEC 60079-11:2011 Explosive atmospheres - Part 11: Equipment protection by intrinsic safety "i"
Edition:6.0

This Certificate **does not** indicate compliance with safety and performance requirements other than those expressly included in the Standards listed above.

TEST & ASSESSMENT REPORTS:

A sample(s) of the equipment listed has successfully met the examination and test requirements as recorded in:

Test Report:

NL/DEK/ExTR24.0044/01

Quality Assessment Reports:

NL/DEK/QAR24.0012/00

US/ETL/QAR20.0010/05



IECEx Certificate of Conformity

Certificate No.: **IECEx DEK 24.0039X**

Page 3 of 4

Date of issue: 2025-07-11

Issue No: 1

EQUIPMENT:

Equipment and systems covered by this Certificate are as follows:

DM-TOX Toxic Gas or Oxygen Deficiency Sensor, for fixed installation, designed to detect and monitor toxic gases or O₂ concentration in air.

Ambient temperature range -40 °C to +55 °C.

PI-VOC Gas Sensor, for fixed installation, designed to detect and monitor a wide range of VOC and toxic gases in air.

Ambient temperature range -20 °C to +55 °C.

SPECIFIC CONDITIONS OF USE: YES as shown below:

1. The sensor shall be installed on a metallic grounded structure in a manner that the final assembly assures that the sensor enclosure is bonded to ground.
2. The sensor's common plane wire (black) shall be connected to ground in the final assembly.
3. Proper precautions shall be taken during installation, operation, maintenance and service of the product in order to avoid the build-up of static charges on the plastic splashguard.
4. The screws holding down the retaining plate are special fasteners of type: stainless steel, Philips Pan-head Machine screws, M3 x 0.5 6g/6H having yield strength greater than 40,000 PSI.
5. The flameproof joints are not intended to be repaired.



IECEx Certificate of Conformity

Certificate No.: **IECEx DEK 24.0039X**

Page 4 of 4

Date of issue: 2025-07-11

Issue No: 1

DETAILS OF CERTIFICATE CHANGES (for issues 1 and above)

Update of the company name for the additional manufacturing location.